

# Flutter Mobile Requirement Audit

Proyek: **QuakeRoute** — quakeroute-mobile | Tanggal: 25 Agustus 2026 | Status: Audit selesai, menunggu konfirmasi implementasi

## 1. Dokumen yang Direview

| #  | Dokumen                       | Ukuran | Peran                                                           |
|----|-------------------------------|--------|-----------------------------------------------------------------|
| 1  | docs/prd.md                   | 40KB   | Tujuan produk, scope MVP, 6 skenario                            |
| 2  | docs/srs.md                   | 44KB   | FR-001–FR-044 testable, NFR, out-of-scope                       |
| 3  | docs/architecture-document.md | 34KB   | Modular monolith, feature-oriented mobile, REST-only            |
| 4  | docs/ui-ux-specification.md   | 51KB   | <b>Prioritas tertinggi UI</b> — screens, komponen, token, state |
| 5  | docs/api-specification.md     | 25KB   | Kontrak REST /api/v1                                            |
| 6  | docs/database-schema.md       | 32KB   | PostgreSQL+PostGIS schema (konteks field)                       |
| 7  | docs/domain-risk-model.md     | 47KB   | Aturan severity/confidence/status, formula cost (backend-owned) |
| 8  | docs/ai-requirement.md        | 43KB   | AI Vision/text extraction (backend-owned)                       |
| 9  | docs/simulation-validation.md | 40KB   | Emergency Simulation + validasi kausal                          |
| 10 | docs/test-cases.md            | 27KB   | Acceptance criteria per fitur                                   |
| 11 | docs/tech-stack.md            | 17KB   | Stack terkunci per teknologi                                    |

Sumber tambahan: quakeroute-api/routes/api.php + modul backend (backend **sudah terimplementasi** dan sesuai kontrak api-specification.md), seluruh quakeroute-mobile/lib/\*\*, pubspec.yaml, .env.example backend.

## 2. Ringkasan Requirement (per Fitur)

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1. **Foundation** — Flutter + Riverpod + dio + flutter\_map + go\_router + geolocator + image\_picker (tech-stack §2); Feature-Oriented Architecture (architecture §5); design tokens ui-ux §3; navigasi hub-and-spoke ui-ux §6.
2. **Dynamic Safety Map** — PRD §9.1, SRS FR-001–FR-004, ui-ux §8.2: peta OSM full-bleed, pin hazard berwarna severity, HUD jumlah hazard aktif, disclaimer, refresh manual/polling.
3. **Destination Selection** — SRS FR-005–FR-007, ui-ux §8.3: daftar shelter/fasilitas medis dari GET /destinations, pilih → request route.
4. **Route Visualization** — SRS FR-008–FR-010, FR-018–FR-020, ui-ux §8.4–8.5: garis rute, strip ringkasan cost breakdown (base + penalty + uncertainty = total), deteksi recalculation (superseded\_by\_route\_id) + banner.
5. **Hazard Reporting** — SRS FR-011–FR-017 (+voice SHOULD-HAVE), ui-ux §8.6–8.9: Photo (AI review + confirm/reject), Text (submit langsung), Quick Tap (6 kategori), Voice kondisional.
6. **Hazard Review** — SRS FR-027–FR-030, ui-ux §8.10–8.11: detail sheet (severity/confidence/status/source/evidence/konflik) + list view.
7. **Emergency Simulation** — SRS FR-036–FR-040, ui-ux §8.12, simulation-validation: pilih skenario, run, bandingkan baseline vs risk-aware, lihat perubahan route saat simulasi berjalan.

## 3. Audit Fitur Detail

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### Fitur 1 — Foundation

- **A. Referensi:** tech-stack §2, architecture-document §5, ui-ux §3, §6.
- **B. Screens:** shell aplikasi + router (8 rute sudah ada di app\_router.dart).
- **C. Flow:** launch → disclaimer (sekali) → Map home.
- **D. Komponen:** QRTokens, tema, ApiClient, model DTO, atom/molecule widgets.
- **E. Interaksi:** navigasi antar screen.
- **F. State UI:** loading app, error network global.
- **G. Data:** konfigurasi .env (API\_BASE\_URL, SESSION\_ID).
- **H. API:** base path /api/v1, header X-Session-Id (placeholder auth TBD).
- **Status:** ☒ SEBAGIAN BESAR TERPENUHI di kode existing.

### Fitur 2 — Dynamic Safety Map

- **A.** SRS FR-001–FR-004; ui-ux §8.2; prd §9.1.
- **B.** MapScreen (home /).
- **C.** Launch → peta → lihat pin hazard → tap pin → detail sheet.
- **D.** QRMapCanvas (ada), marker hazard (belum), HUD chip count (placeholder statis), bottom sheet dinamis (statis), disclaimer overlay (belum ada).
- **E.** Pan/zoom, tap marker, pull-refresh/manual refresh.
- **F.** Loading hazards, empty (0 hazard), error retry, pin per severity (Low→High warna semantic).

- **G.** List hazard: id, type, severity, confidence, road\_impact, status, location(lat/lng), source, timestamp.
- **H.** GET `/hazards?status=&bbox=&updated_since=` — **sudah ada di backend**, mendukung delta-polling via `updated_since`.
- **Status:** ⚠ Skeleton ada; integrasi data belum. **Catatan:** pewarnaan *segmen jalan* tidak mungkin tanpa geometry segmen (lihat GAP-01/GAP-02).

### Fitur 3 — Destination Selection

- **A.** SRS FR-005–FR-007; ui-ux §8.3; api §5.
- **B.** DestinationScreen (`/destinations`).
- **C.** Buka dari map → load list → tap kartu → POST route → navigate ke Route View.
- **D.** QRDestinationCard (nama, tipe badge, jarak, estimasi waktu) — jarak/ETA **tidak tersedia dari API** (lihat GAP-03).
- **E.** Tap pilih destinasi; loading saat generate route.
- **F.** Loading, empty, error retry, submitting.
- **G.** destination\_id, name, type, location.
- **H.** GET `/destinations`, POST `/routes {origin:{lat,lng}, destination_id}` — keduanya **sudah ada di backend** (201 + payload route).
- **Status:** ⚠ Screen kosong (QREmptyState); API siap.

### Fitur 4 — Route Visualization

- **A.** SRS FR-008–FR-010, FR-018–FR-020, FR-034; ui-ux §8.4–8.5; api §6.
- **B.** RoutingScreen (`/route`).
- **C.** Route dibuat → tampil polyline + summary strip → polling GET `/routes/active` → jika `superseded_by_route_id` ≠ null → banner recalculation → tap → fetch route baru.
- **D.** Polyline rute, QRRouteSummaryStrip (total\_cost, base, hazard\_penalty, uncertainty\_penalty), banner “Rute diperbarui”.
- **E.** Tap banner untuk update; dismiss.
- **F.** Loading, no-active-route, error, recalc-banner visible.
- **G.** route\_id, status, supersedes/superseded\_by, total\_cost, segments[road\_segment\_id + 4 field cost].
- **H.** POST `/routes`, GET `/routes/active`, GET `/routes/{id}` — **sudah ada di backend**.
- **Status:** ✖ **BLOCKER** — **GAP-01:** response tidak memuat koordinat/polyline sama sekali (diverifikasi langsung di RouteService.php:101-117, 127-147). Flutter tidak bisa menggambar garis rute hanya dari road\_segment\_id.

### Fitur 5 — Hazard Reporting

- **A.** SRS FR-011–FR-017, FR-035 (voice SHOULD-HAVE); ui-ux §8.6–8.9; api §3.
- **B.** ReportSelector + PhotoReportScreen + TextReportScreen + QuickReportScreen.
- **C. Photo:** capture (image\_picker) → preview + lokasi otomatis → submit multipart → response suggestion PendingConfirmation → QRAIReviewPanel (type/severity/confidence/road\_impact/evidence) → Confirm/Reject → toast.
- **C. Text:** textarea → submit JSON → hazard langsung masuk dataset → konfirmasi sukses.

- **C. Quick:** grid 6 kategori → tap → submit {category, lat, lng} → selesai.
- **D.** Kamera preview, form, AI panel, grid kategori, tombol submit dengan loading.
- **E.** Semua input dikirim apa adanya; **mobile TIDAK mengklasifikasi/mengekstrak apa pun** (boundary terpenuhi oleh desain existing).
- **F.** Permission kamera ditolak, upload progress, AI timeout/fallback, validasi teks kosong, double-submit guard.
- **G.** Foto file + lat/lng; teks + lat/lng; kategori + lat/lng.
- **H.** POST /hazard-reports/photo|text|quick (+/voice), POST /hazard-suggestions/{id}/confirm|reject — **semua sudah ada di backend.**
- **Status:** ⚠ Semua screen masih skeleton; alur lengkap harus dibangun. Lihat CONFLICT-01 untuk mode teks.

## Fitur 6 — Hazard Review

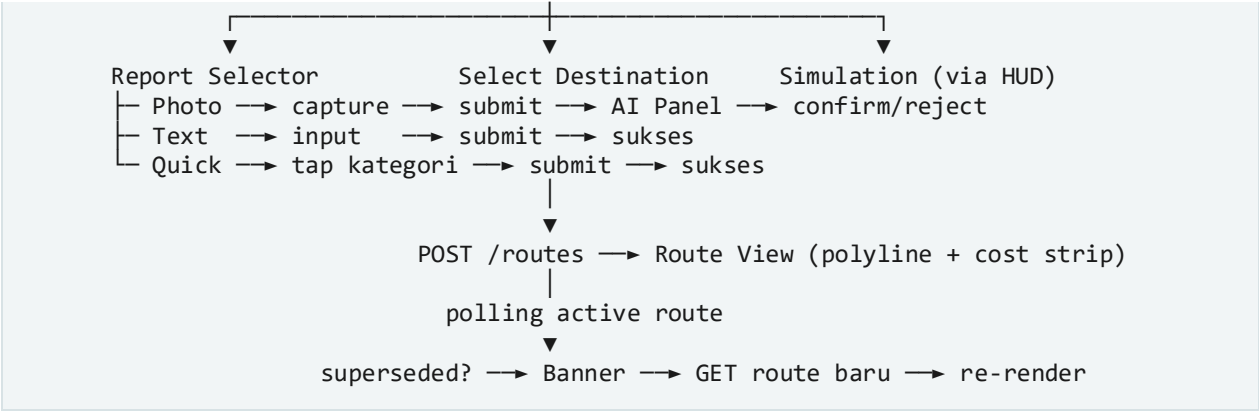
- **A.** SRS FR-027–FR-030; ui-ux §8.10–8.11; api §4.
- **B.** Hazard Detail Sheet (bottom sheet di atas map) + Hazard List Screen.
- **C.** Tap pin/list item → detail → (list) kembali ke map.
- **D.** QRSeverityBadge, QRStatusBadge, QRConfidenceMeter (komponen sudah ada), evidence section, conflict warning.
- **E.** Scroll detail; tutup sheet; filter list (opsional).
- **F.** Loading, not-found, loaded; uncertain/conflicting ditandai visual (never color-alone — badge encode icon+label).
- **G.** Satu hazard lengkap + conflicting\_with[ ].
- **H.** GET /hazards/{id} — ada di backend; list memakai cache hasil GET /hazards.
- **Status:** ✗ Belum ada satupun screen/komponennya — **harus CREATE.**

## Fitur 7 — Emergency Simulation

- **A.** SRS FR-036–FR-040; ui-ux §8.12; simulation-validation §5–§7; api §7.
- **B.** SimulationScreen (/simulation).
- **C.** Load daftar skenario → pilih → run → polling status → hasil: route baseline vs risk-aware + total\_cost comparison → opsi apply-route-simulasi ke map.
- **D.** Kartu skenario, progress indicator, panel perbandingan cost.
- **E.** Pilih skenario, mulai, pantau progres, baca hasil, terapkan rute simulasi.
- **F.** Loading scenarios, running, completed, failed.
- **G.** scenario\_id/name/description, run status, result (baseline vs risk-aware routes + costs).
- **H.** GET /simulation/scenarios, POST /simulation/scenarios/{id}/run, GET /simulation/runs/{run\_id} — **sudah ada di backend.**
- **Status:** ⚠ Skeleton saja.

## 4. User Flow Map (Mobile)

Launch → Disclaimer (sekali) → Dynamic Safety Map (home)



## 5. Data Flow Map

[User input] → Flutter (form/capture/tap, transformasi presentasi saja)  
→ dio POST/GET /api/v1/\* → Laravel (validasi, AI, risk calc, routing engine, simulation)  
→ JSON response → Riverpod state → render UI

Polling loops:

- A) GET /hazards?updated\_since=... → update markers [interval TBD]
- B) GET /routes/active → cek superseded\_by\_route\_id [interval TBD]
- C) GET /simulation/runs/{id} → status Running→Completed [interval TBD]

Larangan boundary: Flutter tidak pernah menghitung severity, confidence, penalty, cost, jalur terpendek, atau klasifikasi AI – semua nilai hanya ditampilkan apa adanya dari response. ✓ Desain existing sudah konsisten.

## 6. API Dependency Audit

| Endpoint               | Dipakai Fitur        | Status Backend | Status Mobile   |
|------------------------|----------------------|----------------|-----------------|
| GET /hazards (+filter) | Map, Review          | ☑ Ada          | ✗ Belum dipakai |
| GET /hazards/{id}      | Review               | ☑ Ada          | ✗               |
| GET /destinations      | Destination          | ☑ Ada          | ✗               |
| POST /routes           | Destination, Routing | ☑ Ada (201)    | ✗               |
| GET /routes/active     | Routing (polling)    | ☑ Ada          | ✗               |

|                                              |                         |                                         |   |
|----------------------------------------------|-------------------------|-----------------------------------------|---|
| GET /routes/{id}                             | Routing                 | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /hazard-reports/photo                   | Reporting               | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /hazard-reports/text                    | Reporting               | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /hazard-reports/quick                   | Reporting               | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /hazard-reports/voice                   | Reporting (SHOULD-HAVE) | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /hazard-suggestions/{id}/confirm reject | Reporting (photo)       | <input checked="" type="checkbox"/> Ada | ✗ |
| GET /simulation/scenarios                    | Simulation              | <input checked="" type="checkbox"/> Ada | ✗ |
| POST /simulation/scenarios/{id}/run          | Simulation              | <input checked="" type="checkbox"/> Ada | ✗ |
| GET /simulation/runs/{run_id}                | Simulation              | <input checked="" type="checkbox"/> Ada | ✗ |

**Kesimpulan:** seluruh dependensi API yang terdokumentasi sudah tersedia di backend; pekerjaan mobile adalah konsumsi murni. Masalahnya bukan endpoint yang hilang, melainkan **field yang hilang dalam response** (lihat Gaps).

## 7. Responsibility Boundary Check

| Tanggung jawab                                                            | Pemilik | Kepatuhan desain existing                                                          |
|---------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------|
| Klasifikasi hazard, ekstraksi AI (foto/teks)                              | Backend | <input checked="" type="checkbox"/> Mobile hanya kirim mentah + tampilkan proposal |
| Severity, confidence, road_impact, status                                 | Backend | <input checked="" type="checkbox"/> DTO read-only (core/models)                    |
| Risk score, hazard_penalty, uncertainty_penalty, segment cost, total_cost | Backend | <input checked="" type="checkbox"/> Field di-parse & ditampilkan, tidak dihitung   |
| Pemilihan jalur, recalculasi                                              | Backend | <input checked="" type="checkbox"/> Mobile hanya poll + render                     |

|                                                                               |         |   |
|-------------------------------------------------------------------------------|---------|---|
| Simulasi skenario & pembanding baseline                                       | Backend | ☑ |
| Render peta, marker, polyline, form, navigasi, transformasi presentasi ringan | Flutter | ☑ |

Tidak ditemukan pelanggaran boundary pada arsitektur existing.

## 8. Requirement Traceability Table

| Flutter Feature | Requirement Source                           | UI Requirement                 | Data Needed       | API Dependency                       | Backend Logic                     | Status                     |
|-----------------|----------------------------------------------|--------------------------------|-------------------|--------------------------------------|-----------------------------------|----------------------------|
| Foundation      | tech-stack §2, arch §5, ui-ux §3/§6          | Shell, tokens, router          | env config        | base /api/v1                         | —                                 | ☑ Ready                    |
| Safety Map      | FR-001–004, ui-ux §8.2, PRD §9.1             | Peta + pin + HUD + disclaimer  | hazard list       | GET /hazards                         | query + filter bbox/delta         | ⚠ Build needed             |
| Destination     | FR-005–007, ui-ux §8.3, api §5               | List kartu destinasi           | destinations      | GET /destinations, POST /routes      | simpan route + supersede lama     | ⚠ Build needed             |
| Route View      | FR-008–010, 018–020, ui-ux §8.4–8.5, api §6  | Polyline + cost strip + banner | route + segments  | GET /routes/active, GET /routes/{id} | routing engine, recalc            | ✗ Blocked GAP-01           |
| Report Photo    | FR-011–014, ui-ux §8.6, api §3.1             | Capture + review panel         | foto + lokasi     | photo + confirm/reject               | AI Vision, suggestion store       | ⚠ Build needed             |
| Report Text     | FR-015–017, ui-ux §8.7<br>⚠konflik, api §3.4 | Form + hasil                   | teks + lokasi     | text                                 | AI text extraction, direct insert | ⚠ Build + resolve conflict |
| Report Quick    | FR-017a, ui-ux §8.8, api §3.3                | Grid 6 kategori                | kategori + lokasi | quick                                | insert langsung, confidence TBD   | ⚠ Build needed             |

|               |                                          |                        |                |                       |                     |                    |
|---------------|------------------------------------------|------------------------|----------------|-----------------------|---------------------|--------------------|
| Report Voice  | FR-035 SHOULD-HAVE, ui-ux §8.9           | Kondisional/omit       | audio          | voice                 | AI voice extraction | ? Butuh keputusan  |
| Hazard Review | FR-027–030, ui-ux §8.10–8.11             | Sheet + list + badges  | hazard detail  | GET /hazards/{id}     | conflict detection  | ✗ Create from zero |
| Simulation    | FR-036–040, ui-ux §8.12, sim-val, api §7 | Picker + run + compare | scenarios/runs | 3 endpoint simulation | simulation engine   | ⚠ Build needed     |

## 9. Gaps

| ID     | Deskripsi                                                                                                                                                                                                                             | Dampak                                                                                                                                                  | Tingkat         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| GAP-01 | Response route (POST /routes, GET /routes/{id}, active) <b>tidak memuat koordinat polyline/geometry segmen</b> — hanya road_segment_id + 4 angka cost (diverifikasi di RouteService.php). Tidak ada endpoint geometry jaringan jalan. | Flutter <b>tidak dapat menggambar garis rute</b> atau mewarnai segmen jalan. FR-008 & sebagian besar fitur Route Visualization mustahil tanpa data ini. | <b>BLOCKING</b> |
| GAP-02 | Tidak ada endpoint yang mengekspos geometry road_segments untuk pewarnaan segmen di Dynamic Safety Map (ui-ux §8.2). Pin titik saja dimungkinkan.                                                                                     | Interpretasi “colored segments” harus diturunkan menjadi “pin hazard” atau backend menambah endpoint geometry.                                          | Perlu keputusan |
| GAP-03 | ui-ux §8.3 kartu destinasi menampilkan jarak & estimasi waktu; GET /destinations tidak menyediakan keduanya. ETA = logika routing (milik backend).                                                                                    | Kartu destinasi tidak lengkap; jarak euclidean bisa dihitung Flutter (transformasi presentasi) tapi ETA tidak boleh.                                    | Perlu keputusan |
| GAP-04 | ui-ux §8.5 strip rute menampilkan durasi tempuh; API tidak punya field durasi/jarak.                                                                                                                                                  | Elemen UI tidak dapat diisi dari data resmi.                                                                                                            | Perlu keputusan |
| GAP-05 | Interval polling (hazards delta, active route, simulation run status) tidak dispesifikasi di dokumen mana pun.                                                                                                                        | Nilai default implementasi harus disepakati (usulan: 15–30 dtk, configurable).                                                                          | Minor           |



|        |                                                                                          |                                                         |      |
|--------|------------------------------------------------------------------------------------------|---------------------------------------------------------|------|
| GAP-06 | Confidence threshold Quick-Tap: docs menyebut “TBD”; backend .env belum memuat nilainya. | Bukan beban mobile (backend yang pakai), hanya dicatat. | Info |
|--------|------------------------------------------------------------------------------------------|---------------------------------------------------------|------|

## 10. Conflicts

| ID          | Konflik                                                                                                                                                                                                                                                                                                                                                | Rekomendasi                                                                                                                                                                                 |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONFLICT-01 | <b>Text report:</b> ui-ux §8.7 mewajibkan QRAIReviewPanel + Confirm/Edit/Reject untuk hasil ekstraksi teks juga, tetapi SRS FR-015–017 & api-specification §3.4 menyatakan teks masuk dataset <b>langsung tanpa langkah konfirmasi</b> (dan memang backend tidak menyediakan suggestion-flow untuk teks). Prioritas ui-ux vs kontrak API bertentangan. | Ikuti SRS/API: teks submit → sukses langsung, tampilkan hasil ekstraksi <b>read-only</b> setelah submit. Mengikuti ui-ux mensyaratkan perubahan backend baru. <b>Perlu konfirmasi user.</b> |
| CONFLICT-02 | Voice: backend sudah punya POST /hazard-reports/voice, sedangkan ui-ux §8.9 bilang jika tidak dibangun, tile Voice <b>dihilangkan total</b> (tanpa tile disabled). Skeleton existing menampilkan tile disabled — melanggar ui-ux.                                                                                                                      | Putuskan bangun voice (butuh dep speech_to_text + izin mic, menambah risiko jadwal) atau hapus tile. <b>Perlu konfirmasi user.</b>                                                          |

## 11. Assumptions

1. X-Session-Id cukup untuk identitas sesi prototype; auth penuh out-of-scope (sesuai PRD/SRS out-of-scope).
2. Emulator Android memakai http://10.0.2.2:8000/api/v1 (default existing ApiClient) — nilai production lewat .env.
3. Polling (bukan push/WebSocket) adalah mekanisme update — konsisten dengan “REST-only” di architecture-document.
4. Jarak straight-line pada kartu destinasi dihitung Flutter (haversine) bila GAP-03 disetujui opsinya.
5. Disclaimer disimpan lokal (mis. shared\_preferences) — perlu 1 dep kecil ATAU flag in-memory per sesi (in-memory = tanpa dep baru; “sekali” menjadi per-launch).
6. Parameter numerik yang masih TBD di docs (bobot severity, decay staleness, dll.) tetap TBD dan **tidak** akan diisi nilai buatan di sisi mobile.

## 12. Penilaian Kode Existing

| File/Area | Keputusan | Catatan |
|-----------|-----------|---------|
|-----------|-----------|---------|

|                                                                   |        |                                                      |
|-------------------------------------------------------------------|--------|------------------------------------------------------|
| pubspec.yaml (deps)                                               | KEEP   | Sudah persis tech-stack §2; tanpa dep berlebih       |
| main.dart, app.dart, app_router.dart                              | KEEP   | 8 rute sesuai ui-ux §6                               |
| core/models/* (DTO)                                               | KEEP   | Mirror akurat api-specification; read-only           |
| core/network/api_client.dart, api_exception.dart                  | KEEP   | Base URL, timeout, mapping error §2 sudah benar      |
| core/state/app_providers.dart                                     | MODIFY | Tambah provider per repository nanti                 |
| core/theme/* (tokens, typography, theme)                          | KEEP   | Token lengkap ui-ux §3                               |
| core/widgets/qr_button/badge/confidence_meter/scaffold/map_canvas | KEEP   | Atom/molecule sesuai §4                              |
| features/map/presentation/map_screen.dart                         | MODIFY | Bottom sheet statis → dinamis; HUD count; disclaimer |
| features/destination/...destination_screen.dart                   | MODIFY | Isi list dari API                                    |
| features/reporting/report_selector_screen.dart                    | MODIFY | Tile Voice disabled → hapus/aktifkan (CONFLICT-02)   |
| features/reporting/photo text quick_tap/...                       | MODIFY | Bangun alur penuh + repository + controller          |
| features/routing/...routing_screen.dart                           | MODIFY | Tergantung resolusi GAP-01                           |

|                                               |               |                                                                                                                                                                              |
|-----------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| features/simulation/...simulation_screen.dart | <b>MODIFY</b> | Bangun picker/run/compare                                                                                                                                                    |
| Belum ada                                     | <b>CREATE</b> | Repository layer per fitur, controller/notifier Riverpod, hazard detail sheet + list screen, safety disclaimer widget, polling service, halaman Settings/About (ui-ux §8.13) |
| —                                             | <b>REMOVE</b> | Tidak ada                                                                                                                                                                    |

### 13. Usulan Rencana Implementasi (Phase 1–7)

| Phase                           | Isi                                                                                                                    | Prasyarat                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>1. Foundation</b>            | Repository layer + Riverpod controllers infra, polling service, disclaimer, Settings/About, error/loading standarisasi | —                                 |
| <b>2. Dynamic Safety Map</b>    | GET /hazards → marker berwarna severity, HUD count, bottom-sheet dinamis, refresh + delta polling                      | Phase 1                           |
| <b>3. Destination Selection</b> | List destinasi, pilih → POST /routes → simpan active route id                                                          | Phase 2                           |
| <b>4. Route Visualization</b>   | Render polyline + cost strip + banner recalculation (polling active)                                                   | Phase 3 + <b>resolusi GAP-01</b>  |
| <b>5. Hazard Reporting</b>      | Photo (multipart + review panel + confirm/reject), Text, Quick (grid + submit), sukses/error states                    | Phase 1 (paralel dgn 2–4 mungkin) |
| <b>6. Hazard Review</b>         | Detail sheet dari pin, list screen, badge status/conflict                                                              | Phase 2                           |
| <b>7. Emergency Simulation</b>  | Scenarios list, run + polling status, panel baseline vs risk-aware                                                     | Phase 4                           |

## 14. Risks

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1. **GAP-01 (blocking)** — tanpa geometry rute, Phase 4 gagal; butuh keputusan: backend menambah field `segments[].geometry / endpoint / road-network`, ATAU mobile merender representasi non-spasial (daftar segmen + cost) yang menurunkan nilai fitur secara signifikan.
2. **CONFLICT-01/02** — keputusan UX teks & voice memengaruhi scope Phase 5.
3. Polling agresif pada device banyak user dapat membebani backend prototype (tanpa cache). Mitigasi: interval configurable + delta `updated_since`.
4. Izin runtime (kamera, lokasi) di Android/iOS perlu penanganan penolakan yang eksplisit (sudah diantisipasi di state F fitur reporting).
5. Jadwal: 6 phase tersisa padat; Phase 5 paling berat (3 sub-alur + multipart + review panel).

## 15. Konfirmasi yang Diperlukan

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1. **GAP-01 (wajib)**: apakah backend akan menambahkan geometry/polyline pada response route (atau endpoint geometry segmen)? Jika ya → Phase 4 penuh; jika tidak → representasi alternatif non-spasial?
2. **GAP-02/03/04**: cukup pin hazard (tanpa pewarnaan segmen) dan kartu destinasi dengan jarak haversine tanpa ETA — setuju?
3. **CONFLICT-01**: mode teks ikuti SRS/API (submit langsung, hasil read-only) — setuju?
4. **CONFLICT-02**: bangun Voice report (tambah dep + izin mic) atau omit total?
5. **GAP-05**: setuju interval polling default 20 detik (configurable via `.env`)?

**Requirement audit selesai. Saya belum melakukan perubahan kode. Saya menunggu konfirmasi sebelum memulai implementasi.**